Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 20 epochs with a batch size of 128. The learning rate was set to 0.001 with the Adam optimizer.

2 Results

Figure 1 shows the f1 score during training.

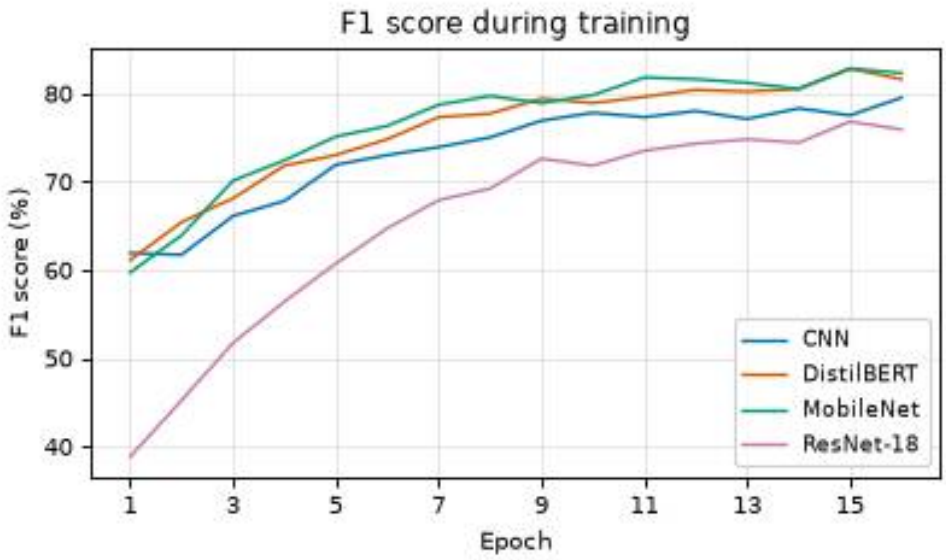


Figure 1: F1 score during training for the models in this experiment.

DistilBERT reaches about 80% F1 score at epoch 7. As shown in Figure 1, from epoch 14 to epoch 15, the F1 score of MobileNet improves by a relative 2.9%. CNN reaches 75.0% F1 score at epoch 8 (Figure 1). As shown in Figure 1, MobileNet outperforms ResNet-18 by 11.2 percentage points at epoch 1.

Figure 2 shows the f1 score during training.

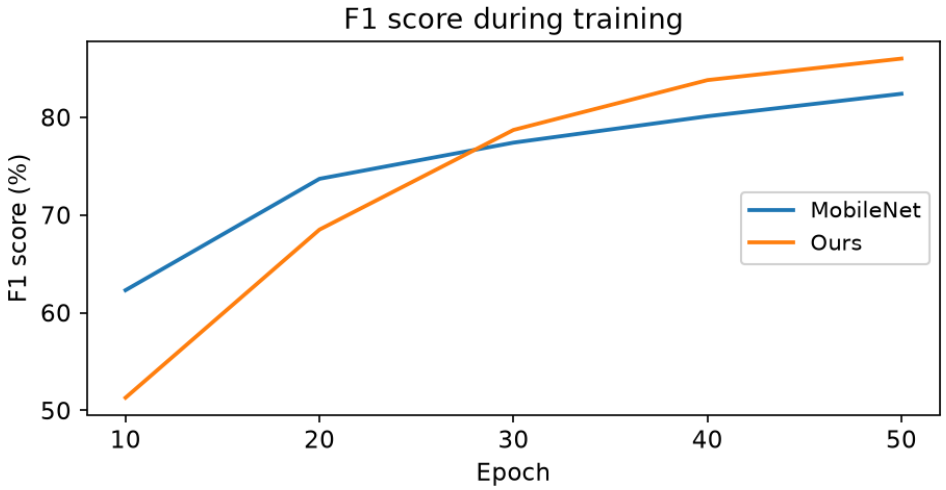


Figure 2: F1 score during training for the models in this experiment.

MobileNet outperforms Ours by 11 percentage points at epoch 10. At epoch 50, Ours is 11.2 points better than MobileNet. From epoch 40 to epoch 50, the F1 score of Ours improves by a relative 1.7% (Figure 2). Ours attains a F1 score of 102.9% at epoch 40.

Figure 3 shows the test accuracy by dataset.

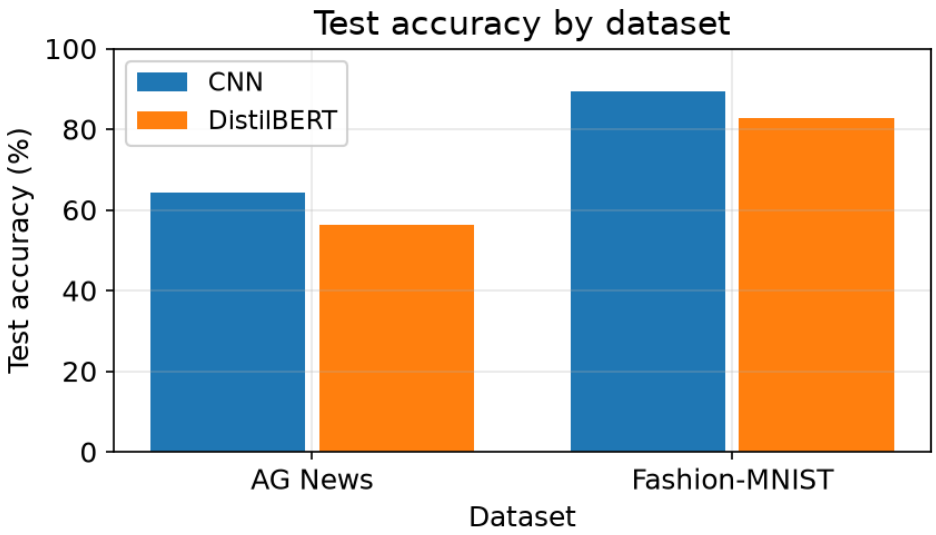


Figure 3: Test accuracy by dataset for the models in this experiment.

As shown in Figure 3, DistilBERT reaches over 80% test accuracy on Fashion-MNIST. CNN outperforms DistilBERT by about 8 percentage points on AG News. DistilBERT reaches about 60% test accuracy on AG News. On AG News, CNN is 18.0 points better than DistilBERT.

3 Discussion

SVM reaches 54.2% accuracy on Yelp. We note that the training was stable for all models.